

Propositional Logic

Solution 1

(a)

$$p \wedge q \Rightarrow p$$

(b)

p	q	$\mathbf{p} \Rightarrow \mathbf{q}$
t	t	t
t	f	f
f	t	t
f	f	t

(c)

$$\neg (p \wedge q) \Leftrightarrow \neg p \vee \neg q$$

Solution 2

(a)

$$(p \vee q) \wedge r \Leftrightarrow (p \wedge r) \vee (q \wedge r)$$

(b)

p	q	$\mathbf{p} \Leftrightarrow \mathbf{q}$
t	t	t
t	f	f
f	t	f
f	f	t

(c)

$$p \Rightarrow (q \Rightarrow r)$$